



ABBOTSLEIGH

Student's Name: _____

Student Number:

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Teacher's Name: _____

2024
HIGHER SCHOOL CERTIFICATE
Assessment 4
Trial Examination

Mathematics Extension 2

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen.
- **NESA approved** calculators may be used.
- **NESA approved** reference sheet is provided.
- All necessary working should be shown in every question.
- Make sure your HSC candidate Number is on the front cover of each booklet.
- Start a new booklet for Each Question.
- Answer the Multiple Choice questions on the answer sheet provided.
- If you do not attempt a whole question, you must still hand in the Writing Booklet, with the words '**NOT ATTEMPTED**' written clearly on the front cover.

Total marks – 100

- Attempt Sections I and II.
- All questions are of equal value.

Section I Pages 3 - 8

10 marks

- Attempt Questions 1–10.
- Allow about 15 minutes for this section.

Section II Pages 9 - 17

90 marks

- Attempt Questions 11– 16.
- Allow about 2 hrs and 45 minutes for this section.

Outcomes to be assessed:

Mathematics Extension 2

HSC :

A student

- | | |
|----------------|---|
| MEX12-1 | understands and uses different representations of numbers and functions to model, prove results and find solutions to problems in a variety of contexts |
| MEX12-2 | chooses appropriate strategies to construct arguments and proofs in both practical and abstract settings |
| MEX12-3 | uses vectors to model and solve problems in two and three dimensions |
| MEX12-4 | uses the relationship between algebraic and geometric representations of complex numbers and complex number techniques to prove results, model and solve problems |
| MEX12-5 | applies techniques of integration to structured and unstructured problems |
| MEX12-6 | uses mechanics to model and solve practical problems |
| MEX12-7 | applies various mathematical techniques and concepts to model and solve structured, unstructured and multi-step problems |
| MEX12-8 | communicates and justifies abstract ideas and relationships using appropriate language, notation and logical argument |

SECTION I

10 marks

Attempt Questions 1 – 10

Use the multiple-choice answer sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

(A) ☐ (B) ☒ (C) ☐ (D) ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

(A) ☒ (B) ☒ (C) ☐ (D) ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows.

(A) ☒ (B) ☒ (C) ☐ (D) ☐

correct ↖

-
1. Consider the statement: “If you understand the chain rule for differentiation it will reduce the number of formulas for derivatives you need to remember”.

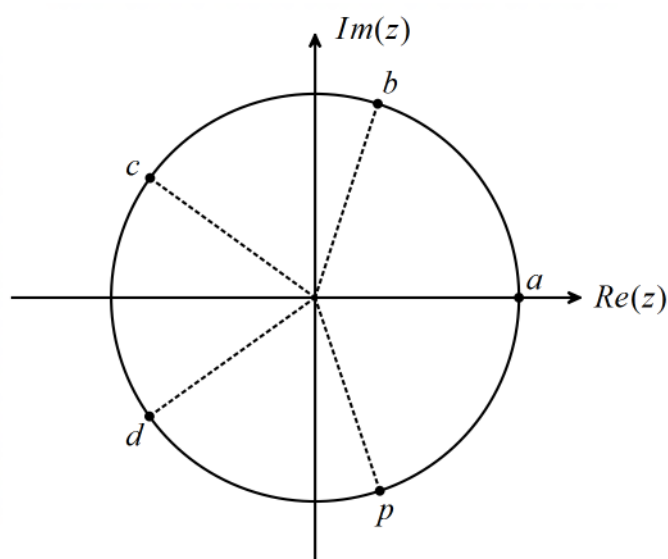
The converse of this statement is:

- A. If you don’t want to remember formulas, understand the chain rule.
- B. If you have reduced the number of formulas for derivatives that you remember, then you understand the chain rule.
- C. If you haven’t reduced the number of derivative formulas you need to remember, you don’t understand the chain rule.
- D. If you don’t understand the chain rule for differentiation you will need to remember a lot of formulas.

2. What is the magnitude of the vector

$$\sin \theta \underline{i} + \cot \theta \underline{j} + \cos \theta \underline{k}, \quad 0 < \theta < \frac{\pi}{2}?$$

- A. 1
- B. $\operatorname{cosec} \theta$
- C. $\cot \theta$
- D. $\sec \theta$
3. If a, b, c, d and p are the fifth roots of unity as indicated on the diagram below, which of the points represents the square root of p ?



- A. a
- B. b
- C. c
- D. d

4. Which of the following is a correct expression for $\int \frac{2x+1}{\sqrt{9-x^2}} dx$

A. $\sin^{-1}\left(\frac{x}{3}\right) - \sqrt{9-x^2} + C$

B. $\cos^{-1}\left(\frac{x}{3}\right) - 2\sqrt{9-x^2} + C$

C. $\sin^{-1}\left(\frac{x}{3}\right) - 2\sqrt{9-x^2} + C$

D. $-\cos^{-1}\left(\frac{x}{3}\right) - \sqrt{9-x^2} + C$

5. z satisfies the equation $|z - \sqrt{3} + i| = 1$. The minimum value of $\arg(z)$ is

A. 0°

B. -30°

C. -60°

D. -75°

6. $OACB$ is a rectangle with $\overrightarrow{OA} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} -1 \\ 4 \\ b \end{pmatrix}$.

Which of the following is the vector $\overrightarrow{OC}$?

A. $\begin{pmatrix} 2 \\ 4 \\ 1 \end{pmatrix}$

B. $\begin{pmatrix} 2 \\ 4 \\ 5 \end{pmatrix}$

C. $\begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}$

D. $\begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix}$

7. Which of the following is a symbolic representation of the statement:

“At least one of your friends is perfect”

Let $P(x)$ be “ x is perfect” and let $F(x)$ be “ x is your friend” and let the domain be all people.

A. $\forall x (F(x) \rightarrow P(x))$

B. $\forall x (F(x) \cap P(x))$

C. $\exists x (F(x) \cap P(x))$

D. $\exists x (F(x) \rightarrow P(x))$

8. A particle of mass 1 kg is projected vertically upwards with a speed u from the origin O . The particle is subject to a constant gravitational force and a resistance that is proportional to half the square of its velocity $v \text{ ms}^{-1}$, the constant of proportionality being k .

Let x be the displacement in metres of the particle above O at time t seconds after the particle is projected and g be the acceleration due to gravity.

Which of the following expressions gives the maximum height reached by the particle?

A.
$$\int_u^0 \frac{dv}{-g - \frac{k}{2}v^2}$$

B.
$$\int_u^0 \frac{v \, dv}{-g - \frac{k}{2}v^2}$$

C.
$$\int_u^0 \frac{dv}{-g + \frac{k}{2}v^2}$$

D.
$$\int_u^0 \frac{v \, dv}{-g + \frac{k}{2}v^2}$$

9. What is the value of $\int_1^e x^3 \ln(x) \, dx$

A. $\frac{3e^2}{2} - 1$

B. $\frac{3e^2}{2} + 1$

C. $\frac{1}{16}(3e^4 - 1)$

D. $\frac{1}{16}(3e^4 + 1)$

10. Consider the spheres:

$$S_1 : (x+1)^2 + (y-3)^2 + (z+2)^2 = 4$$

$$S_2 : (x-1)^2 + (y+1)^2 + (z-2)^2 = r^2$$

If these spheres are tangential to each other, what are the possible values of r ?

A. $r = 2$ or 4

B. $r = 4$ or 8

C. $r = 2$ or 8

D. $r = 4$ or 6

End of Section I

SECTION II

Total Marks – 90

Attempt Questions 11 - 16

All questions are of equal value

Answer each question in a **SEPARATE** writing booklet. Extra writing booklets are available.

In Questions 11-16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet. **Marks**

(a) Solve: $z^2 + z + 3 = 0$, $z \in \mathbb{C}$, giving your answers in Cartesian form. **2**

(b) Find the equation of the line that is parallel to $r(\lambda) = (2\hat{i} + 3\hat{j} - \hat{k}) + \lambda(-\hat{i} + 2\hat{j} - 4\hat{k})$
that passes through the point $\begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$. **2**

(c) Prove that $n^3 - n$ is divisible by 6. **3**

(d) By considering Euler's representation of a complex number, find the exact value of

$$\sqrt{-1}^{\sqrt{-1}}. \quad \text{2}$$

(e) Prove that the sum of two irrational numbers is an irrational number or disprove by counter-example. **2**

(f) Find $\int \frac{x+16}{x^2+2x-8} dx$ **4**

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) Prove that $\log_2 5$ is irrational.

3

(b) Find

$$\int_0^{\pi} \left(\frac{\cot \theta + 1}{\operatorname{cosec} \theta} - \frac{\sec \theta}{\tan \theta + \cot \theta} \right) d\theta.$$

3

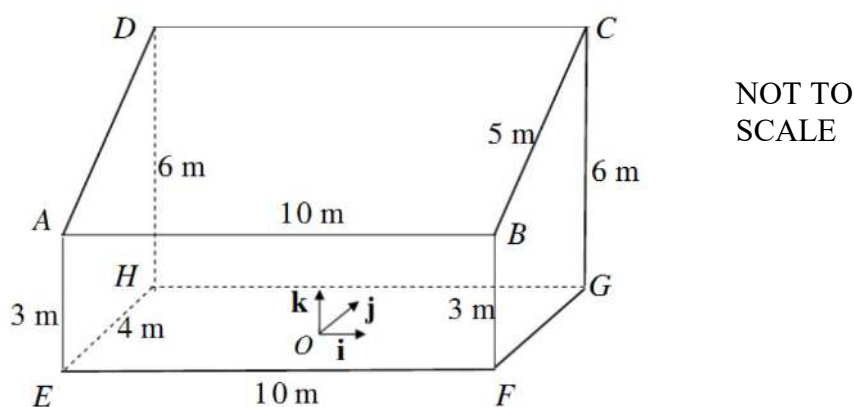
(c) Given $i(\omega + 2) = z$ and $\omega z = -4i$ where $\operatorname{Im}(\omega) < 0$

3

Find the exact values of w and z in the form $a + ib$ where a and b are real.

Question 12 continues on page 11

- (d) The diagram below shows a schematic of the design of a garage.



$ABCD$ is an inclined rectangular roof, where $AB = 10$ m and $AD = 5$ m. $EFGH$ is a rectangle on horizontal ground, where $EF = 10$ m and $EH = 4$ m. The points A and B are 3 m directly above E and F respectively. The points C and D are 6 m directly above G and H respectively.

The point O is at the centre of $EFGH$ and is taken to be the origin. Perpendicular unit vectors $\underline{i}$, $\underline{j}$ and $\underline{k}$ are in the directions of $\overrightarrow{EF}$, $\overrightarrow{EH}$ and $\overrightarrow{EA}$ respectively.

- (i) Show that $\overrightarrow{AC} = \begin{pmatrix} 10 \\ 4 \\ 3 \end{pmatrix}$ and find a vector equation of the line passing through A and C . 2

- (ii) A spotlight is erected on a vertical pillar at Q with coordinates $\begin{pmatrix} 15 \\ 6 \\ 0 \end{pmatrix}$. 1
- Find the height of the pillar if the spotlight is collinear with A and C .

- (iii) The main electrical supply for the garage runs along the line between A and C . 3
- Find the coordinates of the point, X , on AC such that the distance from a light mounted at D to the main electric supply is minimised.

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) Find $\int \ln(x + \sqrt{x^2 - 4}) dx$ **3**

(b) A series, $u_1 + u_2 + u_3 + \dots$ has terms defined by:

$$u_1 = 2 \text{ and } u_{r+1} = u_r + \frac{1}{2^r}$$

Prove by mathematical induction that $u_n = 3 - \left(\frac{1}{2}\right)^{n-1}$ for all $n \in \mathbb{Z}^+$. **3**

(c) The acceleration of a particle moving along the x – axis is given by: **4**

$$a = 2x^3 - 5x \text{ ms}^{-2}$$

If the particle has velocity, $v = 2 \text{ ms}^{-2}$ when $x = 0$, find an expression for v in terms of x and determine what values x can take.

(d) Given the polynomial $P(x) = x^4 - 2x^2 - 3x + a$ has a factor of $x^2 + x + 1$, find the value of a and hence find the roots of $P(x)$. **3**

(e) Let $\underline{u} = \underline{i} + \underline{j}$ and $\underline{v} = \underline{i} + 2\underline{k}$. Find the unit vector that is perpendicular to $\underline{u}$ and $\underline{v}$. **2**

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) Find an expression for $\int \frac{2x \ln(2x^2 + 1)}{2x^2 + 1} dx$ **3**

(b) Given $a, b, x, y, z \in \mathbb{R}^+$

(i) Prove the AM - GM inequality **1**

$$\frac{a+b}{2} \geq \sqrt{ab}$$

(ii) Further, if $x + y + z = 1$, prove **3**

$$\frac{xy}{z} + \frac{yz}{x} + \frac{zx}{y} \geq 1$$

(iii) Given $\frac{a+b+c+d}{4} \geq \sqrt[4]{abcd}$, $a, b, c, d \in \mathbb{R}^+$, prove **3**

$$\frac{a+b+c}{3} \geq \sqrt[3]{abc}$$

(c) A cube has diagonally opposite vertices at the origin, O, and $A(2a, 4a, 6a)$.

(i) Prove that the vertices of the cube lie on the sphere with equation

$$x^2 + y^2 + z^2 = 2ax + 4ay + 6az$$

Note: The edges of the cube are not parallel to the axes. **4**

(ii) Find the coordinates of the point on the sphere that has the greatest negative x -value. **1**

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

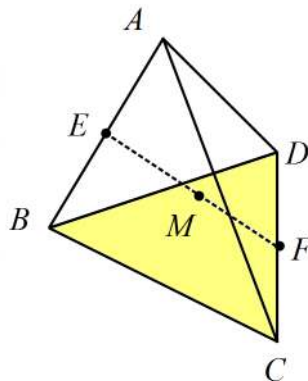
Marks

(a) For $n \geq 0$, define: $I_n = \int_0^{\frac{\pi}{4}} \sec^n x \, dx$

(i) Show that $I_n = \frac{(\sqrt{2})^{n-2}}{n-1} + \frac{n-2}{n-1} I_{n-2}$ for $n \geq 2$, **3**

(ii) Hence evaluate $\int_0^{\frac{\pi}{4}} \sec^6 x \, dx$. **2**

- (b) Consider the tetrahedron shown with vertices A, B, C and D . Let each vertex be represented by the position vector $\underline{a}, \underline{b}, \underline{c}$ and $\underline{d}$ respectively. (i.e. $\overrightarrow{OA} = \underline{a}$ etc).



M is the midpoint of the line $\overrightarrow{EF}$ where E and F are the midpoints of AB and CD respectively. **4**
Find an expression for $\overrightarrow{OM}$ in terms of the position vectors $\underline{a}, \underline{b}, \underline{c}$ and $\underline{d}$.

Question 15 continues on page 15

Question 15 (continued)**Marks**

- (c) (i) Given $P_n(x) = (x - a_1)(x - a_2) \dots (x - a_n)$ prove by mathematical induction that **3**

$$\frac{P'_n(x)}{P_n(x)} = \frac{1}{(x - a_1)} + \frac{1}{(x - a_2)} + \dots + \frac{1}{(x - a_n)} \quad \text{for } n \geq 2.$$

- (ii) Consider ω is an n th root of unity. **3**

$$\text{Let } P(z) = (z - \omega)(z - \omega^2) \dots (z - \omega^{n-1}).$$

By finding another expression for $P(z)$ or otherwise, show that

$$\frac{1}{1 - \omega} + \frac{1}{1 - \omega^2} + \dots + \frac{1}{1 - \omega^{n-1}} = \frac{n-1}{2}$$

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) Given $x \geq 0$

(i) Prove that $1 - x \leq \frac{1}{1+x} \leq 1$. **2**

(ii) For $x > 0$, show that $1 - \frac{1}{2x} \leq x \ln \left(1 + \frac{1}{x} \right) \leq 1$. **3**

(iii) Hence deduce that $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x = e$. **2**

Question 16 continues on page 17.

Question 16 (continued).**Marks**

- (b) An object of mass 1 kg is dropped from a height, h metres above the ground. There is a constant gravitational force of g and a resistive force of kv newtons, where $v \text{ ms}^{-1}$ is the velocity of the object. Hence $a = g - kv$.

(i) Show that the velocity of the particle at time t is $v = \frac{g}{k} (1 - e^{-kt})$ 2

- (ii) Show that the position of the object below the point of release at time t is

$$x = \frac{gt}{k} + \frac{g}{k^2} (e^{-kt} - 1) \quad 2$$

At the same time the object is released a second, identical object, is projected vertically upward from the ground with velocity $u \text{ ms}^{-1}$. Taking the origin as the ground and the upward direction as positive, the velocity of the second particle is given by:

$$v = \frac{g + ku}{k} e^{-kt} - \frac{g}{k} \quad (\text{DO NOT PROVE THIS})$$

- (iii) Show that the position, X , of the second particle below h is given by 2

$$X = h + \frac{g + ku}{k^2} (e^{-kt} - 1) + \frac{g}{k} t$$

- (iv) If the two objects collide at time T , show $T = \frac{1}{k} \ln \left(\frac{u}{u - kh} \right)$ 2

End of paper

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2024 EXT 2 TRIAL SOLUTIONS

- ① Converse: "To reduce the number of formulas you need to remember, you need to understand the chain rule."

B

$$\begin{aligned} \textcircled{2} |\sin \theta \underline{x} + \cot \theta \underline{j} + \csc \theta \underline{k}| &= \sqrt{\sin^2 \theta + \cot^2 \theta + \csc^2 \theta} \\ &= \sqrt{1 + \cot^2 \theta} \\ &= \sqrt{\csc^2 \theta} \\ &= \csc \theta \end{aligned}$$

B

- ③ To square a unit vector double its arg.

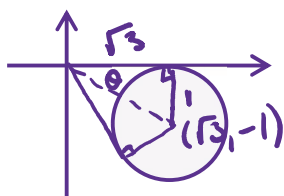
$$\therefore a^2 = a, b^2 = c, c^2 = p, d^2 = b, p^2 = d$$

C

$$\begin{aligned} \textcircled{4} \int \frac{2x+1}{\sqrt{9-x^2}} dx &= - \int \frac{-2x-1}{\sqrt{9-x^2}} dx \\ &= - \left[\int \frac{-2x}{\sqrt{9-x^2}} dx - \int \frac{1}{\sqrt{9-x^2}} dx \right] \\ &= - \frac{(9-x^2)^{1/2}}{1/2} + \sin^{-1}\left(\frac{x}{3}\right) + C \\ &= \sin^{-1}\left(\frac{x}{3}\right) - 2\sqrt{9-x^2} + C \end{aligned}$$

C

- ⑤ $|z - (\sqrt{3} - i)| = 1 \Rightarrow$ circle, centre $(\sqrt{3} - i)$
radius = 1



$$\min \arg(z) = -2\theta$$

$$\tan \theta = \frac{1}{\sqrt{3}}$$

$$\theta = 30^\circ$$

$$\therefore \min \arg(z) = -60$$

C

- ⑥ if $OACB$ is a rectangle, then $\vec{OA} \cdot \vec{OB} = 0$
as angles in a rectangle are 90°



$$\therefore \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 4 \\ b \end{pmatrix} = 0$$

$$\text{i.e. } -2 - 4 + 3b = 0$$

$$3b = 6$$

$$b = 2$$

$$\vec{OC} = \vec{OA} + \vec{OB}$$

$$= \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + \begin{pmatrix} -1 \\ 4 \\ 2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix}$$

⑦

$$\exists F(x) \wedge P(x) \Rightarrow \exists x (F(x) \wedge P(x))$$

C

⑧ $\ddot{x} = -g - \frac{1}{2}kv^2$

$$v \frac{dv}{dx} = -g - \frac{1}{2}kv^2$$

$$\int_0^H \frac{v dv}{-g - \frac{1}{2}kv^2} = \int_0^H dx \quad \text{where } H \text{ is max. height.}$$

B

⑨ $I = \int_1^e x^3 \ln x \, dx$

let $u = \ln x$

$$u' = \frac{1}{x}$$

$$v' = x^3$$

$$v = \frac{x^4}{4}$$

$$= \left[\frac{x^4}{4} \ln x \right]_1^e - \int_1^e \frac{x^3}{4} dx$$

$$= \frac{e^4}{4} - \left[\frac{x^4}{16} \right]_1^e$$

$$= \frac{e^4}{4} - \left(\frac{e^4}{16} - \frac{1}{16} \right)$$

$$= \frac{3e^4}{16} + \frac{1}{16}$$

⑩

- ⑩ Let D be the distance between the centres of the spheres.

$$\begin{aligned} d &= \sqrt{(1-1)^2 + (-3-1)^2 + (2-2)^2} \\ &= \sqrt{4+16+16} \\ &= 6 \end{aligned}$$

For S_1 & S_2 to touch externally, the sum of their radii must equal 6

$$\begin{aligned} \therefore r + 2 &= 6 \\ r &= 4 \end{aligned}$$

For S_1 & S_2 to touch internally, the difference of their radii must equal 6

$$\begin{aligned} r - 2 &= 6 \\ r &= 8 \end{aligned}$$

B.

SECTION II

QUESTION 11

a) $z^2 + z + 3 = 0 \quad z \in \mathbb{C}$

$$z^2 + z + \frac{1}{4} = -3 + \frac{1}{4} \quad \textcircled{1}$$

$$(z + \frac{1}{2})^2 = -\frac{11}{4}$$

$$z + \frac{1}{2} = \pm \frac{\sqrt{11}i}{2}$$

$$z = -\frac{1}{2} \pm \frac{\sqrt{11}i}{2} \quad \textcircled{1}$$

b) $r(\lambda) = (2\hat{i} + 3\hat{j} - \hat{k}) + \lambda(-\hat{i} + 2\hat{j} - 4\hat{k})$

Direction vector $(-\hat{i} + 2\hat{j} - 4\hat{k}) \therefore$ line // through $(2, -1, 3)$

$$p(\lambda) = (2\hat{i} - \hat{j} + 3\hat{k}) + \lambda(-\hat{i} + 2\hat{j} - 4\hat{k})$$

①

①

c) Prove $n^3 - n$ is divisible by 6

$$\begin{aligned} n^3 - n &= n(n^2 - 1) \\ &= n(n-1)(n+1) \\ &= (n-1)n(n+1) \end{aligned} \quad (1)$$

Hence, $n^3 - n$ is the product of 3 consecutive numbers. So, one of the numbers must be (1) divisible by 3 & (at least) one of the numbers is even so must be divisible by 2

$\Rightarrow n^3 - n$ is divisible by 2 & 3 (1)

$\Rightarrow n^3 - n$ is divisible by 6.

d) $F_1 = i$
 $= e^{i\frac{\pi}{2}}$ (1)

$$\begin{aligned} \text{so } F_1^{F_1} &= (e^{i\pi/2})^i \\ &= e^{i^2 \frac{\pi}{2}} \\ &= e^{-\pi/2} \quad (\text{a real number}) \end{aligned} \quad (1)$$

e) consider $2+\sqrt{3}$ & $2-\sqrt{3}$, both irrational (1)

then $2+\sqrt{3} + 2-\sqrt{3} = 4$, a rational number

$\therefore$ The sum of two irrational numbers is not necessarily irrational. (1)

$$\begin{aligned} \text{f) } I &= \int \frac{x+16}{x^2+2x-8} dx \\ &= \int \frac{x+16}{(x+4)(x-2)} dx \end{aligned} \quad (1)$$

Consider $\frac{x+16}{(x+4)(x-2)} = \frac{A}{x+4} + \frac{B}{x-2}$
 i.e. $x+16 = A(x-2) + B(x+4)$

Let $x = 2$: $18 = 6B$
 $B = 3$
 Let $x = -4$: $12 = -6A$
 $A = -2$

} ①

$\therefore I = \int \left(\frac{-2}{x+4} + \frac{3}{x-2} \right) dx$ ①
 $= -2 \ln |x+4| + 3 \ln |x-2| + C$
 $= \ln \left[\frac{17x-21^3}{(x+4)^2} \right] + C$ ①

QUESTION 12

a) Assume that $\log_2 5$ is rational

i.e. let $\log_2 5 = \frac{p}{q}$ where p & q are mutually prime integers > 0 ①

so $5 = 2^{p/q}$ From definition of logs

or $5^q = 2^p$ ①

but all powers of 5 are odd & all powers of 2 are even ①

$\therefore 5^q \neq 2^p$

so $\log_2 5$ cannot be rational (proof by contradiction)

$$b) I = \int_0^{\pi} \left(\frac{\cot\theta + 1}{\csc\theta} - \frac{\sec\theta}{\tan\theta \cot\theta} \right) d\theta$$

$$\begin{aligned} \text{now } \frac{\cot\theta + 1}{\csc\theta} - \frac{\sec\theta}{\tan\theta \cot\theta} &= \frac{\frac{\cos\theta}{\sin\theta} + 1}{\frac{1}{\sin\theta}} - \frac{\frac{1}{\cos\theta}}{\frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta}} \quad (1) \\ &= \sin\theta \left(\frac{\cos\theta}{\sin\theta} + 1 \right) - \frac{1}{\cos\theta} \left(\frac{\sin^2\theta + \cos^2\theta}{\sin\theta \cos\theta} \right) \\ &= \cos\theta + \sin\theta - \frac{1}{\cos\theta} \left(\frac{\cos\theta \sin\theta}{\cos^2\theta + \sin^2\theta} \right) \\ &= \cos\theta + \sin\theta - \sin\theta \quad (\text{as } \cos^2\theta + \sin^2\theta = 1) \\ &= \cos\theta \quad (1) \end{aligned}$$

$$\begin{aligned} \therefore I &= \int_0^{\pi} \cos\theta \, d\theta \\ &= [\sin\theta]_0^{\pi} \\ &= 0 \quad (1) \end{aligned}$$

$$c) z = i(\omega + 2) \quad (1)$$

$$\omega z = -4i \quad (2)$$

Sub (1) in (2)

$$\omega(i(\omega + 2)) = -4i$$

$$\omega^2 + 2\omega + 1 = -4 + 1$$

$$(\omega + 1)^2 = -3$$

$$\omega + 1 = \pm \sqrt{3}i \quad (1)$$

$$\omega = -1 \pm \sqrt{3}i$$

$$\text{Given } \text{Im}(\omega) < 0 \Rightarrow \omega = -1 - \sqrt{3}i \quad (1)$$

$$\text{Sub in (1): } z = i(-1 - \sqrt{3}i + 2)$$

$$= i(1 - \sqrt{3}i)$$

$$= \sqrt{3} + i \quad (1)$$

d) i) for the given position of O

$$\vec{OA} = -5\hat{i} - 2\hat{j} + 3\hat{k}$$

$$\vec{OC} = 5\hat{i} + 2\hat{j} + 6\hat{k}$$

$$\vec{AC} = \vec{OC} - \vec{OA}$$

$$= 5\hat{i} + 2\hat{j} + 6\hat{k} - (-5\hat{i} - 2\hat{j} + 3\hat{k}) \quad (1)$$

$$= 10\hat{i} + 4\hat{j} + 3\hat{k} \quad \text{as required.}$$

$\therefore$ Equation of AC

$$\underline{r}(\lambda) = -5\underline{i} - 2\underline{j} + 3\underline{k} + \lambda(10\underline{i} + 4\underline{j} + 3\underline{k}) \quad (1)$$

(ii) We need to find λ such that $\underline{r}(\lambda) = 15\underline{i} + 6\underline{j} + h\underline{k}$
where h is the height of the pillar.

Consider $\underline{i}$ component:

$$-5\underline{i} + 10\lambda\underline{i} = 15\underline{i}$$

$$\text{i.e.} \quad 10\lambda = 20$$

$$\lambda = 2$$

$$\therefore \underline{r}(2) = 15\underline{i} + 6\underline{j} + (3 + 2(3))\underline{k}$$

$$= 15\underline{i} + 6\underline{j} + 9\underline{k} \quad (1)$$

$\therefore$ height of pillar is 9m

$$\text{iii) } \vec{OB} = -5\hat{i} + 2\hat{j} + 6\hat{k}$$

$$\vec{AB} = \vec{OB} - \vec{OA}$$

$$= 4\hat{j} + 3\hat{k} \quad (1)$$

To find x , find projection of $\vec{AB}$ on $\vec{AC}$

$$\begin{aligned}\text{Proj}_{\vec{n}} \vec{a} &= \frac{(4\vec{j} + 3\vec{k}) \cdot (10\vec{i} + 4\vec{j} + 3\vec{k})}{(10\vec{i} + 4\vec{j} + 3\vec{k}) \cdot (10\vec{i} + 4\vec{j} + 3\vec{k})} (10\vec{i} + 4\vec{j} + 3\vec{k}) \\ &= \frac{16+9}{100+16+9} (10\vec{i} + 4\vec{j} + 3\vec{k}) \\ &= \frac{1}{5} (10\vec{i} + 4\vec{j} + 3\vec{k}) \quad (1)\end{aligned}$$

$$\begin{aligned}\therefore \vec{D_K} &= (-5\vec{i} - 2\vec{j} + 3\vec{k}) + \frac{1}{5}(10\vec{i} + 4\vec{j} + 3\vec{k}) \\ &= -3\vec{i} - \frac{6}{5}\vec{j} + \frac{18}{5}\vec{k} \quad (1)\end{aligned}$$

QUESTION 13

$$a) I = \int \ln(x + \sqrt{x^2 - 4}) dx$$

$$\text{let } u = \ln(x + \sqrt{x^2 - 4})$$

$$u' = 1$$

$$u' = \frac{1}{x + \sqrt{x^2 - 4}} \cdot (1 + \frac{1}{2}(x^2 - 4)^{-1/2} \cdot 2x)$$

$$v = x$$

$$= \frac{1}{x + \sqrt{x^2 - 4}} \cdot \left(1 + \frac{x}{\sqrt{x^2 - 4}}\right)$$

$$= \frac{1}{x + \sqrt{x^2 - 4}} \left(\frac{\sqrt{x^2 - 4} + x}{\sqrt{x^2 - 4}} \right) \quad (1)$$

$$= \frac{1}{\sqrt{x^2 - 4}}$$

$$\therefore I = x \ln(x + \sqrt{x^2 - 4}) - \int \frac{x}{\sqrt{x^2 - 4}} dx \quad (1)$$

$$= x \ln(x + \sqrt{x^2 - 4}) - \frac{1}{2} \int \frac{2x}{\sqrt{x^2 - 4}} dx$$

$$= x \ln(x + \sqrt{x^2 - 4}) - \frac{1}{2} \frac{(x^2 - 4)^{1/2}}{1/2} + C$$

$$= x \ln(x + \sqrt{x^2 - 4}) - \sqrt{x^2 - 4} + C \quad (1)$$

b) $u_1 = 2$ $u_{r+1} = u_r + \frac{1}{2^r}$
 Prove $u_n = 3 - (\frac{1}{2})^{n-1}$ for $n \in \mathbb{Z}^+$

Step 1: Prove true for $n=1$

$$\begin{aligned} u_1 &= 2 \text{ (given)} & u_1 &= 3 - (\frac{1}{2})^{1-1} \\ & & &= 3 - 1 \\ & & &= 2 \end{aligned} \quad \textcircled{1}$$

$\therefore$ true for $n=1$

Step 2: assume true for $n=k$

i.e. $u_k = 3 - (\frac{1}{2})^{k-1}$

Step 3: Prove true for $n=k+1$ given true for $n=k$

i.e. RTP $u_{k+1} = 3 - (\frac{1}{2})^k$

$$\begin{aligned} u_{k+1} &= u_k + \frac{1}{2^k} \text{ from definition.} \\ &= 3 - (\frac{1}{2})^{k-1} + \frac{1}{2^k} \text{ by assumption } \textcircled{1} \\ &= 3 - \left[\frac{1}{2^{k-1}} - \frac{1}{2^k} \right] \\ &= 3 - \left[\frac{1}{2^{k-1}} \left(1 - \frac{1}{2} \right) \right] \\ &= 3 - \left[\frac{1}{2^{k-1}} \cdot \frac{1}{2} \right] \\ &= 3 - \frac{1}{2^k} \\ &= 3 - (\frac{1}{2})^k \text{ as required. } \textcircled{1} \end{aligned}$$

Step 4: So the result holds by the inductive process.

c) $a = 2x^3 - 5x$

$$\frac{d}{dx}(\pm v^2) = 2x^3 - 5x$$

$$\pm v^2 = \frac{2x^4}{4} - \frac{5x^2}{2} + C$$

$$v^2 = x^4 - 5x^2 + 2C$$

①

when $x=0$, $v=2$.

$$\therefore 2^2 = 0 - 0 + 2C$$

$$2C = 4$$

$$\therefore v^2 = x^4 - 5x^2 + 4$$

①

$$v = \pm \sqrt{x^4 - 5x^2 + 4}$$

Since $v=2$ when $x=0$, take positive root

$$\text{so } v = \sqrt{x^4 - 5x^2 + 4}$$

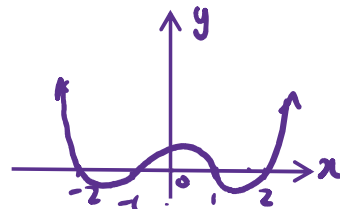
①

For domain: $x^4 - 5x^2 + 4 \geq 0$

$$(x^2 - 1)(x^2 - 4) \geq 0$$

critical values: $x^2 = 1, 2$

$$x = \pm 1, \pm 2$$



as $x=0$ is part of the domain.

$$-1 \leq x \leq 1$$

①

d) $p(x) = x^4 - 2x^2 - 3x + a$

$$\begin{array}{r} x^2 - x - 2 \\ x^2 + x + 1 \overline{) x^4 + 0x^3 - 2x^2 - 3x + a} \\ \underline{x^4 + x^3 + x^2} \end{array}$$

$$\begin{array}{r} -x^3 - 3x^2 - 3x \\ -x^3 - x^2 - x \end{array}$$

$$\underline{-2x^2 - 2x + a}$$

$$\underline{-2x^2 - 2x - 2}$$

$$a + 2$$

①

as x^2+x+1 is a factor, the remainder is 0

$$\therefore a+2=0$$

$$a = -2$$

①

$$\text{So } p(x) = x^4 - 2x^2 - 3x - 2$$

$$= (x^2+x+1)(x^2-x-2)$$

$$= (x^2+x+1)(x-2)(x+1)$$

①

$\therefore$ for roots $x = 2, -1$ or $x^2+x+1=0$

$$x^2+x+\frac{1}{4} = -1+\frac{1}{4}$$

$$(x+\frac{1}{2})^2 = -\frac{3}{4}$$

$$x+\frac{1}{2} = \pm \frac{\sqrt{3}i}{2}$$

$$x = -\frac{1}{2} \pm \frac{\sqrt{3}i}{2}$$

$\therefore$ roots: $x = -1, 2, \frac{-1 \pm \sqrt{3}i}{2}$

①

e) let $\underline{w} = a\underline{i} + b\underline{j} + c\underline{k}$ be $\perp$ to $\underline{u} \notin \underline{v}$

$$\therefore \underline{u} \cdot \underline{w} = 0$$

$$\text{or } \underline{v} \cdot \underline{w} = 0$$

$$(i+j)(a\underline{i}+b\underline{j}+c\underline{k})=0$$

$$(i+2k) \cdot (a\underline{i}+b\underline{j}+c\underline{k})=0$$

$$a+b=0$$

$$a+2c=0$$

$$b = -a$$

$$c = -\frac{a}{2}$$

$$\therefore \underline{w} = a\underline{i} - a\underline{j} - \frac{a}{2}\underline{k}$$

$$= \frac{a}{2}(2\underline{i} - 2\underline{j} - \underline{k})$$

$$= \lambda(2\underline{i} - 2\underline{j} - \underline{k})$$

$$\underline{\hat{w}} = \frac{\lambda(2\underline{i} - 2\underline{j} - \underline{k})}{\sqrt{(2\lambda)^2 + (-2\lambda)^2 + (\lambda)^2}}$$

$$= \frac{\lambda(2\underline{i} - 2\underline{j} - \underline{k})}{\sqrt{9\lambda^2}}$$

$$= \frac{1}{3}(2\underline{i} - 2\underline{j} - \underline{k})$$

①

QUESTION 14

$$a) \quad I = \int \frac{2x \ln(2x^2+1)}{2x^2+1} dx$$

$$\text{let } u = \ln(2x^2+1)$$

$$\frac{du}{dx} = \frac{1}{2x^2+1} \times 4x$$

$$= \frac{4x}{2x^2+1}$$

$$du = \frac{4x}{2x^2+1} dx$$

①

$$\therefore I = \frac{1}{2} \int \frac{4x \ln(2x^2+1)}{2x^2+1} dx$$

$$= \frac{1}{2} \int u du$$

①

$$= \frac{1}{2} \times \frac{u^2}{2} + C$$

$$= \frac{1}{4} (\ln(2x^2+1))^2 + C$$

①

b) (i) for $a, b \geq 0$, we have

$$(\sqrt{a} - \sqrt{b})^2 \geq 0$$

$$a - 2\sqrt{ab} + b \geq 0$$

$$a + b \geq 2\sqrt{ab}$$

$$\frac{a+b}{2} \geq \sqrt{ab} \quad \#$$

①

$$(ii) \quad x+y+z=1$$

$$\text{consider } \frac{xy}{2} + \frac{yz}{2} \geq 2\sqrt{\frac{xy \cdot yz}{x^2}}$$

$$= 2y$$

①

①

$$\text{Similarly } \frac{xy}{2} + \frac{zx}{2} \geq 2x$$

②

$$\& \quad \frac{yz}{2} + \frac{zx}{2} \geq 2z$$

③

} ①

adding ①, ② + ③ we have:

$$\frac{xy}{z} + \frac{yz}{x} + \frac{zx}{y} + \frac{yz}{x} + \frac{zx}{y} + \frac{xy}{z} \geq 2y + 2x + 2z$$

$$2\left(\frac{xy}{z} + \frac{yz}{x} + \frac{zx}{y}\right) \geq 2(x+y+z) \quad \text{①}$$

$$\frac{xy}{z} + \frac{yz}{x} + \frac{zx}{y} \geq 1 \quad \text{as } x+y+z=1$$

(iii) Given $\frac{a+b+c+d}{4} \geq \sqrt[4]{abcd}$ (★)

let $a=a, b=b, c=c, d=\frac{a+b+c}{3}$ ①

Sub in (★):

$$\frac{a+b+c+\frac{a+b+c}{3}}{4} \geq \left[abc\left(\frac{a+b+c}{3}\right)\right]^{1/4} \quad \text{①}$$

$$\frac{\frac{3a+3b+3c+a+b+c}{3}}{4} \geq \left[abc\left(\frac{a+b+c}{3}\right)\right]^{1/4}$$

$$\frac{a+b+c}{3} \geq (abc)^{1/4} \left(\frac{a+b+c}{3}\right)^{1/4}$$

$$\left(\frac{a+b+c}{3}\right)^{3/4} \geq (abc)^{1/4} \quad \text{as } a, b, c > 0$$

$$\left[\left(\frac{a+b+c}{3}\right)^{3/4}\right]^{4/3} \geq \left[(abc)^{1/4}\right]^{4/3} \quad \text{①}$$

$$\frac{a+b+c}{3} \geq \sqrt[3]{abc}$$

#

OR $\therefore a+b+c = 3d$

sub in (★):

$$\frac{3d+d}{4} \geq \sqrt[4]{abcd}$$

$$d \geq (abcd)^{1/4}$$

$$d^{3/4} \geq (abc)^{1/4}$$

$$d \geq (abc)^{1/3}$$

$$\frac{a+b+c}{3} \geq \sqrt[3]{abc}$$

①
①
① } OR

c) (i) If the vertices of the cube lie on a sphere, the centre of the sphere must be the midpoint of OA:

$$M = \left(\frac{0+2a}{2}, \frac{0+4a}{2}, \frac{0+6a}{2} \right) \\ = (a, 2a, 3a) \quad \textcircled{1}$$

$$\text{Radius} = |\vec{OM}| \\ = \sqrt{a^2 + (2a)^2 + (3a)^2} \\ = a\sqrt{14} \quad \textcircled{1}$$

$$\therefore \text{Sphere is: } \left| \begin{pmatrix} x \\ y \\ z \end{pmatrix} - \begin{pmatrix} a \\ 2a \\ 3a \end{pmatrix} \right| = a\sqrt{14} \quad \textcircled{1}$$

$$\therefore \sqrt{(x-a)^2 + (y-2a)^2 + (z-3a)^2} = a\sqrt{14}$$

$$x^2 - 2ax + a^2 + y^2 - 4ay + 4a^2 + z^2 - 6az + 9a^2 = 14a^2$$

$$x^2 + y^2 + z^2 - 2ax - 4ay - 6az + 14a^2 = 14a^2$$

$$\text{i.e. } x^2 + y^2 + z^2 = 2ax + 4ay + 6az$$

(ii) Greatest negative λ -value will occur at the centre minus the radius in the x -direction

$$\text{i.e. } \rho = \begin{pmatrix} a \\ 2a \\ 3a \end{pmatrix} - \begin{pmatrix} \sqrt{14}a \\ 0 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} a(1-\sqrt{14}) \\ 2a \\ 3a \end{pmatrix} \quad \textcircled{1}$$

QUESTION 15

a) $I_n = \int_0^{\pi/4} \sec^n x \, dx, \quad n \geq 0$

(i) $I_n = \int_0^{\pi/4} (\sec^{n-2} x \cdot \sec^2 x) \, dx$

Let $u = \sec^{n-2} x$

$v' = \sec^2 x$

$u' = (n-2)(\sec^{n-3} x)(\sec x \tan x)$

$v = \tan x$

$\therefore I_n = \left[\sec^{n-2} x \tan x \right]_0^{\pi/4} - (n-2) \int_0^{\pi/4} \sec^{n-2} x \tan^2 x \, dx$

$= (\sqrt{2})^{n-2} \cdot 1 - 0 - (n-2) \int_0^{\pi/4} \sec^{n-2} x (\sec^2 x - 1) \, dx$

$= \sqrt{2}^{n-2} - (n-2) [I_n - I_{n-2}]$

$\therefore I_n + (n-2)I_n = \sqrt{2}^{n-2} + (n-2)I_{n-2}$

$(n-1)I_n = \sqrt{2}^{n-2} + (n-2)I_{n-2}$

$I_n = \frac{\sqrt{2}^{n-2}}{n-1} + \frac{n-2}{n-1} I_{n-2} \quad \neq$

(ii) $I_6 = \int_0^{\pi/4} \sec^6 x \, dx$

$= \frac{(\sqrt{2})^4}{5} + \frac{4}{5} I_4$

$= \frac{4}{5} + \frac{4}{5} \left(\frac{(\sqrt{2})^2}{3} + \frac{2}{3} I_2 \right)$

$= \frac{4}{5} + \frac{8}{15} + \frac{8}{15} \left(\frac{\sqrt{2}^0}{1} + 0 \right)$

$= \frac{12+8+8}{15}$

$= \frac{28}{15}$

$$b) \quad \overrightarrow{AB} = \underline{b} - \underline{a}$$

$$\begin{aligned}\overrightarrow{OE} &= \overrightarrow{OA} + \frac{1}{2}\overrightarrow{AB} \\ &= \underline{a} + \frac{1}{2}(\underline{b} - \underline{a}) \\ &= \frac{1}{2}(\underline{a} + \underline{b}) \quad \textcircled{1}\end{aligned}$$

$$\overrightarrow{CO} = \underline{d} - \underline{c}$$

$$\begin{aligned}\overrightarrow{OF} &= \overrightarrow{OC} + \frac{1}{2}\overrightarrow{CO} \\ &= \underline{c} + \frac{1}{2}(\underline{d} - \underline{c}) \\ &= \frac{1}{2}(\underline{c} + \underline{d}) \quad \textcircled{1}\end{aligned}$$

$$\begin{aligned}\overrightarrow{OM} &= \frac{1}{2}(\overrightarrow{OE} + \overrightarrow{OF}) \quad \textcircled{1} \quad \text{or} \quad \overrightarrow{EF} = \overrightarrow{OF} - \overrightarrow{OE} \\ &= \frac{1}{2} \left[\frac{1}{2}(\underline{a} + \underline{b}) + \frac{1}{2}(\underline{c} + \underline{d}) \right] \\ &= \frac{1}{4}(\underline{a} + \underline{b} + \underline{c} + \underline{d}) \quad \textcircled{1} \\ &= \frac{1}{2}(\underline{c} + \underline{d}) - \frac{1}{2}(\underline{a} + \underline{b}) \\ &= \frac{1}{2}(\underline{c} + \underline{d} - \underline{a} - \underline{b}) \quad \textcircled{1}\end{aligned}$$

$$\begin{aligned}\overrightarrow{OM} &= \overrightarrow{OE} + \frac{1}{2}\overrightarrow{EF} \\ &= \frac{1}{2}(\underline{a} + \underline{b}) + \frac{1}{4}(\underline{c} + \underline{d} - \underline{a} - \underline{b}) \\ &= \frac{1}{4}(\underline{a} + \underline{b} + \underline{c} + \underline{d}) \quad \textcircled{1}\end{aligned}$$

$$c) \text{ii) } P_n(x) = (x-a_1)(x-a_2) \dots (x-a_n)$$

$$\text{Prove } \frac{P'_n(x)}{P_n(x)} = \frac{1}{x-a_1} + \frac{1}{x-a_2} + \dots + \frac{1}{x-a_n} \quad n \geq 2$$

Step 1: Prove for $n=2$

$$P_2(x) = (x-a_1)(x-a_2)$$

$$P'_2(x) = (x-a_1) + (x-a_2) \quad (\text{Product rule})$$

$$\begin{aligned}\therefore \frac{P'_2(x)}{P_2(x)} &= \frac{(x-a_1) + (x-a_2)}{(x-a_1)(x-a_2)} \\ &= \frac{x-a_1}{(x-a_1)(x-a_2)} + \frac{x-a_2}{(x-a_1)(x-a_2)} \quad \textcircled{1} \\ &= \frac{1}{x-a_2} + \frac{1}{x-a_1} \quad \therefore \text{true for } n=1\end{aligned}$$

Step 2: assume true for $n=k$

$$\text{i.e. } \frac{P'_k(x)}{P_k(x)} = \frac{1}{x-a_1} + \frac{1}{x-a_2} + \dots + \frac{1}{x-a_k}$$

Step 3: prove true for $n=k+1$ if true for $n=k$

$$\text{RTP } \frac{P'_{k+1}(x)}{P_{k+1}(x)} = \frac{1}{x-a_1} + \frac{1}{x-a_2} + \dots + \frac{1}{x-a_{k+1}}$$

$$\begin{aligned} \text{Now } P_{k+1}(x) &= (x-a_1)(x-a_2) \dots (x-a_k)(x-a_{k+1}) \\ &= P_k(x)(x-a_{k+1}) \end{aligned}$$

$$P'_{k+1}(x) = P_k(x) \cdot 1 + (x-a_{k+1}) P'_k(x)$$

$$\therefore \frac{P'_{k+1}(x)}{P_{k+1}(x)} = \frac{P_k(x) + (x-a_{k+1}) P'_k(x)}{P_{k+1}(x)} \quad (1)$$

$$= \frac{P_k(x)}{P_k(x)(x-a_{k+1})} + \frac{(x-a_{k+1}) P'_k(x)}{P_k(x)(x-a_{k+1})}$$

$$= \frac{1}{x-a_{k+1}} + \frac{P'_k(x)}{P_k(x)}$$

$$= \frac{1}{x-a_{k+1}} + \frac{1}{x-a_1} + \frac{1}{x-a_2} + \dots + \frac{1}{x-a_k} \quad (1)$$

from assumption

= RHS

$$(ii) \quad P(z) = (z-\omega)(z-\omega^2) \dots (z-\omega^{n-1})$$

$$\text{now } z^n - 1 = (z-1)(z^{n-1} + z^{n-2} + \dots + z + 1)$$

$$\text{So } \frac{z^n - 1}{z - 1} = (z^{n-1} + z^{n-2} + \dots + z + 1)$$

also, since $1, \omega, \omega^2, \dots, \omega^{n-1}$ are the roots of $z^n - 1$

$$z^n - 1 = (z-1)(z-\omega)(z-\omega^2) \dots (z-\omega^{n-1}) \quad (1)$$

$$\text{or } \frac{z^n - 1}{z - 1} = (z - \omega)(z - \omega^2) \dots (z - \omega^{n-1})$$

$$\therefore P(z) = 1 + z + z^2 + \dots + z^{n-1}$$

$$\frac{P'(z)}{P(z)} = \frac{1}{z - \omega} + \frac{1}{z - \omega^2} + \dots + \frac{1}{z - \omega^{n-1}} \quad \text{from (i)}$$

note

$$\frac{P'(1)}{P(1)} = \frac{1}{1 - \omega} + \frac{1}{1 - \omega^2} + \dots + \frac{1}{1 - \omega^{n-1}} \quad (\star) \text{ (1)}$$

we also have:

$$P(z) = 1 + z + \dots + z^{n-1}$$

$$\text{with } P'(z) = 0 + 1 + 2z + 3z^2 + \dots + (n-1)z^{n-2}$$

$$= 1 + z + 2z^2 + \dots + (n-1)z^{n-2}$$

$$\text{Here: } P(1) = 1 + 1 + 1^2 + \dots + 1^{n-1}$$

$$= n \times 1$$

$$= n \quad \text{(1)}$$

$$\text{A } P'(1) = 1 + 2 + 3 + \dots + (n-1)$$

$$= \frac{n-1}{2} (1 + (n-1)) \quad \text{Sum of A.P.}$$

$$= \frac{n(n-1)}{2} \quad \text{(1)}$$

$$\therefore \frac{P'(1)}{P(1)} = \frac{n(n-1)}{2} \div n$$

$$= \frac{n-1}{2}$$

$$= \frac{1}{1 - \omega} + \frac{1}{1 - \omega^2} + \dots + \frac{1}{1 - \omega^{n-1}} \quad \text{from } (\star)$$

QUESTION 16

a) (i) $x \geq 0$

$$\therefore -x^2 \leq 0 \leq x \quad (1)$$

$$1-x^2 \leq 1 \leq 1+x$$

since $x \geq 0$, $1+x > 0$ $\therefore$ can divide by $1+x$

$$\frac{1-x^2}{1+x} \leq \frac{1}{1+x} \leq 1 \quad (1)$$

$$1-x \leq \frac{1}{1+x} \leq 1 \quad \text{as required.}$$

(ii) Integrating the result in (i)

$$\int_0^x (1-x) dx \leq \int_0^x \frac{1}{1+x} dx \leq \int_0^x 1 dx \quad \text{for } x > 0$$

$$\left[x - \frac{x^2}{2} \right]_0^x \leq [\ln|1+x|]_0^x \leq [x]_0^x \quad (1)$$

$$x - \frac{x^2}{2} \leq \ln|1+x| \leq x$$

$$\div x: \quad 1 - \frac{x}{2} \leq \frac{1}{x} \ln|1+x| \leq 1 \quad (1)$$

$$\text{now let } x = \frac{1}{n} \quad 1 - \frac{1}{2n} \leq \frac{1}{n} \ln\left(1 + \frac{1}{n}\right) \leq 1 \quad (1)$$

$$1 - \frac{1}{2n} \leq n \ln\left(1 + \frac{1}{n}\right) \leq 1 \quad (\text{also value not required as } x > 0)$$

as n is just a dummy variable, we could write

$$1 - \frac{1}{2x} \leq x \ln\left(1 + \frac{1}{x}\right) \leq 1$$

(iii) we can rewrite the result from (ii) as

$$1 - \frac{1}{2x} \leq \ln\left(1 + \frac{1}{x}\right)^x \leq 1$$

raising e :

$$e^{1-1/2x} \leq e^{\ln\left(1 + \frac{1}{x}\right)^x} \leq e \quad (1)$$

$$e^{1-1/2x} \leq \left(1 + \frac{1}{x}\right)^x \leq e$$

$$\text{taking } \lim_{x \rightarrow \infty} \quad \lim_{x \rightarrow \infty} e^{1-1/2x} \leq \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x \leq \lim_{x \rightarrow \infty} e$$

$$\lim_{x \rightarrow \infty} \frac{1}{2x} = 0 \quad \Rightarrow \quad e^1 \leq \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x \leq e \quad (1)$$

$\therefore \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$ as it is sandwiched.

$$b) i) a = g - kv$$

$$\frac{dv}{dt} = g - kv$$

$$\int_0^v \frac{dv}{g - kv} = \int_0^t dt$$

$$-\frac{1}{k} \int_0^v \frac{-k dv}{g - kv} = [t]_0^t \quad (1)$$

$$-\frac{1}{k} [\ln |g - kv|]_0^v = t$$

$$\ln |g - kv| - \ln g = -kt$$

$$\ln \left| \frac{g - kv}{g} \right| = -kt$$

$$\frac{g - kv}{g} = e^{-kt}$$

$$g - kv = ge^{-kt} \quad (1)$$

$$kv = g - ge^{-kt}$$

$$v = \frac{g}{k} (1 - e^{-kt}) \quad \#$$

$$ii) \text{ so } \frac{d\lambda}{dt} = \frac{g}{k} (1 - e^{-kt})$$

$$\int_0^\lambda d\lambda = \frac{g}{k} \int_0^t (1 - e^{-kt}) dt$$

$$\lambda = \frac{g}{k} \left[t + \frac{1}{k} e^{-kt} \right]_0^t$$

$$= \frac{g}{k} \left[t + \frac{1}{k} e^{-kt} - (0 + \frac{1}{k} e^0) \right] \quad (1)$$

$$= \frac{g}{k} \left[t + \frac{1}{k} e^{-kt} - \frac{1}{k} \right]$$

$$= \frac{gt}{k} + \frac{g}{k^2} (e^{-kt} - 1) \quad \# \quad (1)$$

$$(iii) \quad v = \frac{g+ku}{k} e^{-kt} - \frac{g}{k}$$

$$\text{so } \frac{dx}{dt} = \frac{g+ku}{k} e^{-kt} - \frac{g}{k}$$

$$\int_0^x dx = \int_0^t \left(\frac{g+ku}{k} e^{-kt} - \frac{g}{k} \right)$$

$$x = \left[\frac{g+ku}{k} \cdot \left(-\frac{1}{k}\right) e^{-kt} - \frac{gt}{k} \right]_0^t \quad (1)$$

$$= -\frac{g+ku}{k^2} e^{-kt} - \frac{gt}{k} - \left(-\frac{g+ku}{k^2} e^0 - 0 \right)$$

$$= \frac{g+ku}{k^2} (1 - e^{-kt}) - \frac{gt}{k} \quad (1)$$

$$\text{so } x = h - x$$

$$= h - \left[\frac{g+ku}{k^2} (1 - e^{-kt}) - \frac{gt}{k} \right]$$

$$= h + \frac{g+ku}{k^2} (e^{-kt} - 1) + \frac{gt}{k} \quad \#$$

(iv) For collision $x = X$ where x is position of object 1

$$\therefore \frac{gt}{k} + \frac{g}{k^2} (e^{-kt} - 1) = h + \frac{g+ku}{k^2} (e^{-kt} - 1) + \frac{gt}{k}$$

$$(e^{-kt} - 1) \left[\frac{g}{k^2} - \frac{g+ku}{k^2} \right] = h \quad (1)$$

$$(e^{-kt} - 1) \left(-\frac{u}{k} \right) = h$$

$$e^{-kt} - 1 = -\frac{hk}{u}$$

$$e^{-kt} = 1 - \frac{hk}{u}$$

$$e^{-kt} = \frac{u - hk}{u}$$

$$e^{kt} = \frac{u}{u - hk}$$

$$kt = \ln \left| \frac{u}{u - hk} \right| \quad (1)$$

$$t = \frac{1}{k} \ln \left| \frac{u}{u - hk} \right| \quad \#$$